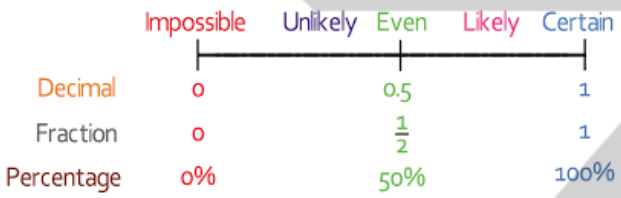
# 1. Probability

## 1.1. Introduction to Probability

- Probability is the study of chance or the likelihood of an event happening

$$P(event) = \frac{\text{Number of favourable outcomes}}{\text{Total number of outcomes}}$$

- If probability = 0, the event is impossible
- If probability =1, the event is certain to happen
- If probability = 0.5, there is an even chance
- All probabilities lie between 0 and 1
- Probability can be written in
  - Fraction
  - Decimal
  - Percentage

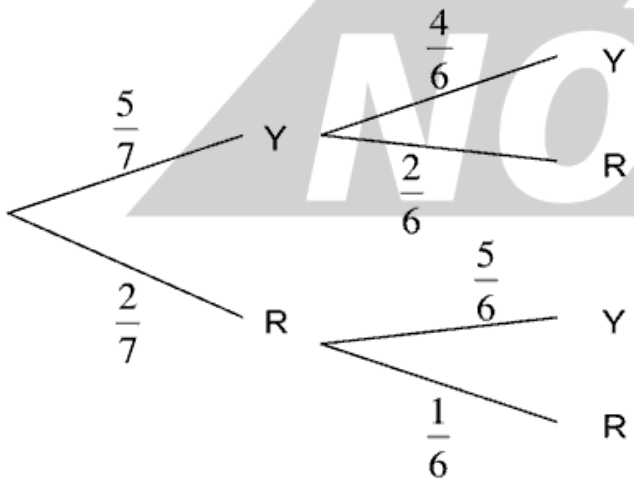



- $P(A) = 1 - P(A')$
- $\text{Probability of A happening} = 1 - \text{Probability of A Not Happening}$

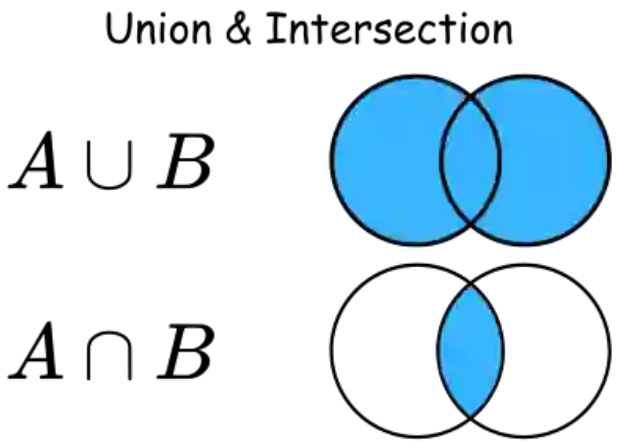
## 1.2. Events

### Exclusive Events

- Two events are exclusive if they cannot occur at the same time




Probabilities of any choice of events




### Independent Events

- Two events are independent if the occurrence of one is unaffected by occurrence of other

### The OR Rule

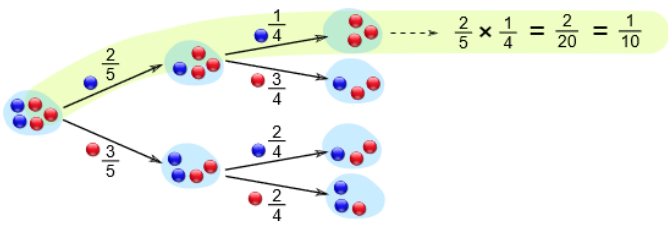
- Refers to the probability of either event happening (or both)
- **Rule:** $P(A \text{ or } B) = P(A) + P(B) - P(A \cap B)$
- If events are mutually exclusive:
  - $P(A \text{ or } B) = P(A) + P(B)$

### The AND Rule

- Refers to the probability of two events happening together.
- **Rule:** $P(A \text{ and } B) = P(A \cap B)$
- If events are independent:
  - $P(A \cap B) = P(A) \times P(B)$

## 1.3. Conditional Probability

- Calculate using Venn diagram:
- Construct the Venn diagram, using sample space of both events
  - $P(A \mid B) = \frac{P(A \cap B)}{P(B)}$
- Calculate using tree diagrams:




- First branches = 1st event outcomes
- Second branches = 2nd event outcomes
- Write probabilities next to each branch
- Multiply along branches for combined probabilities
- Add if multiple paths lead to the desired outcome



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- Note: The probabilities reduce with each step if objects are not replaced

Calculate using two-way tables:

- Column and row headers are the sample space of the two events
  - Fill in each cell with the correct number of outcomes
  - Take the required number from the table and divide by the sum of all values in the row/column of the condition provided.

## 1.4. Relative & Expected Frequencies

### Expected Frequencies

- This is the number of times we would expect an event to occur over a given number of trials that take place during an experiment

$$\text{Expected Frequency} = \text{Probability of Event} \times \text{Number of Trials}$$

### Relative Frequencies

- Relative frequency is an estimate of the **probability** of an event based on **experimental (or observed) results** rather than theoretical probability.
- $\text{Relative Frequency} = \frac{\text{Number of times event occurs}}{\text{total number of trials}}$
- **Example**

  If you toss a coin 100 times and it lands on heads 55 times, the **relative frequency** of getting heads is: $\frac{55}{100} = 0.55$



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